

Assure DeFi™

The Verification **Gold Standard**™



Security Assessment **SHILLD Token**

August 30, 2023

Audit Status: Pass

Audit Edition: Advance



ASSURE DEFI™
THE VERIFICATION **GOLD STANDARD**

Risk Analysis

Classifications of Manual Risk Results

Classification	Description
 Critical	Danger or Potential Problems.
 High	Be Careful or Fail test.
 Low	Pass, Not-Detected or Safe Item.
 Informational	Function Detected

Manual Code Review Risk Results

Contract Privilege	Description
 Buy Tax	1 %
 Sale Tax	1 %
 Cannot Sale	Pass
 Cannot Sale	Pass
 Max Tax	5 %
 Modify Tax	No
 Fee Check	Pass
 Is Honeypot?	Not Detected
 Trading Cooldown	Not Detected
 Can Pause Trade?	Not Detected
 Pause Transfer?	Not Detected
 Max Tx?	Pass
 Is Anti Whale?	Not Detected
 Is Anti Bot?	Not Detected

Contract Privilege	Description
 Is Blacklist?	Not Detected
 Blacklist Check	Pass
 is Whitelist?	Not Detected
 Can Mint?	Pass
 Is Proxy?	Not Detected
 Can Take Ownership?	Not Detected
 Hidden Owner?	Not Detected
 Owner	0x931E6be91f6c0453Cdd4B036D0d9d696F1fAACB7
 Self Destruct?	Not Detected
 External Call?	Not Detected
 Other?	Not Detected
 Holders	1
 Auditor Confidence	Medium

The following quick summary it's added to the project overview; however, there are more details about the audit and its results. Please read every detail.

Project Overview

Token Summary

Parameter	Result
Address	0x3adf095d04311d7BF05C838F0d3DC34a83D81aB6
Name	SHILLD
Token Tracker	SHILLD (SHILLD)
Decimals	18
Supply	10,000,000
Platform	Ethereum
compiler	v0.8.9+commit.e5eed63a
Contract Name	SHILLD
Optimization	Yes with 200 runs
LicenseType	Unlicensed
Language	Solidity
Codebase	https://etherscan.io/address/0x3adf095d04311d7BF05C838F0d3DC34a83D81aB6#code
Payment Tx	Corporate

Main Contract Assessed Contract Name

Name	Contract	Live
SHILLD	0x3adf095d04311d7BF05C838F0d3DC34a83D81aB6	Yes

TestNet Contract Assessed Contract Name

Name	Contract	Live
SHILLD	0xf543a843fa842977cb31c0c4cd31a80ff7c06164	Yes

Solidity Code Provided

SolidID	File Sha-1	FileName
SHILLD	13c3cea6384260e2b8b016a650d516f7ac8048d9	SHILLD.sol
SHILLD		
SHILLD		
SHILLD		

Smart Contract Vulnerability Checks

The Smart Contract Weakness Classification Registry (SWC Registry) is an implementation of the weakness classification scheme proposed in EIP-1470. It is loosely aligned to the terminologies and structure used in the Common Weakness Enumeration (CWE) while overlaying a wide range of weakness variants that are specific to smart contracts.

ID	Severity	Name	File	location
SWC-100	Pass	Function Default Visibility	SHILLD.sol	L: 0 C: 0
SWC-101	Pass	Integer Overflow and Underflow.	SHILLD.sol	L: 0 C: 0
SWC-102	Pass	Outdated Compiler Version file.	SHILLD.sol	L: 0 C: 0
SWC-103	Low	A floating pragma is set.	SHILLD.sol	L: 2 C: 0
SWC-104	Pass	Unchecked Call Return Value.	SHILLD.sol	L: 0 C: 0
SWC-105	Pass	Unprotected Ether Withdrawal.	SHILLD.sol	L: 0 C: 0
SWC-106	Pass	Unprotected SELFDESTRUCT Instruction	SHILLD.sol	L: 0 C: 0
SWC-107	Pass	Read of persistent state following external call.	SHILLD.sol	L: 0 C: 0
SWC-108	Low	State variable visibility is not set..	SHILLD.sol	L: 407 C: 33, L: 432 C: 9
SWC-109	Pass	Uninitialized Storage Pointer.	SHILLD.sol	L: 0 C: 0
SWC-110	Pass	Assert Violation.	SHILLD.sol	L: 0 C: 0
SWC-111	Pass	Use of Deprecated Solidity Functions.	SHILLD.sol	L: 0 C: 0
SWC-112	Pass	Delegate Call to Untrusted Callee.	SHILLD.sol	L: 0 C: 0
SWC-113	Pass	Multiple calls are executed in the same transaction.	SHILLD.sol	L: 0 C: 0

ID	Severity	Name	File	location
SWC-114	Pass	Transaction Order Dependence.	SHILLD.sol	L: 0 C: 0
SWC-115	Pass	Authorization through tx.origin.	SHILLD.sol	L: 0 C: 0
SWC-116	Pass	A control flow decision is made based on The block.timestamp environment variable.	SHILLD.sol	L: 0 C: 0
SWC-117	Pass	Signature Malleability.	SHILLD.sol	L: 0 C: 0
SWC-118	Pass	Incorrect Constructor Name.	SHILLD.sol	L: 0 C: 0
SWC-119	Pass	Shadowing State Variables.	SHILLD.sol	L: 0 C: 0
SWC-120	Pass	Potential use of block.number as source of randomness.	SHILLD.sol	L: 0 C: 0
SWC-121	Pass	Missing Protection against Signature Replay Attacks.	SHILLD.sol	L: 0 C: 0
SWC-122	Pass	Lack of Proper Signature Verification.	SHILLD.sol	L: 0 C: 0
SWC-123	Pass	Requirement Violation.	SHILLD.sol	L: 0 C: 0
SWC-124	Pass	Write to Arbitrary Storage Location.	SHILLD.sol	L: 0 C: 0
SWC-125	Pass	Incorrect Inheritance Order.	SHILLD.sol	L: 0 C: 0
SWC-126	Pass	Insufficient Gas Griefing.	SHILLD.sol	L: 0 C: 0
SWC-127	Pass	Arbitrary Jump with Function Type Variable.	SHILLD.sol	L: 0 C: 0
SWC-128	Pass	DoS With Block Gas Limit.	SHILLD.sol	L: 0 C: 0
SWC-129	Pass	Typographical Error.	SHILLD.sol	L: 0 C: 0
SWC-130	Pass	Right-To-Left-Override control character (U+202E).	SHILLD.sol	L: 0 C: 0
SWC-131	Pass	Presence of unused variables.	SHILLD.sol	L: 0 C: 0

ID	Severity	Name	File	location
SWC-132	Pass	Unexpected Ether balance.	SHILLD.sol	L: 0 C: 0
SWC-133	Pass	Hash Collisions with Multiple Variable Length Arguments.	SHILLD.sol	L: 0 C: 0
SWC-134	Pass	Message call with hardcoded gas amount.	SHILLD.sol	L: 0 C: 0
SWC-135	Pass	Code With No Effects (Irrelevant/Dead Code).	SHILLD.sol	L: 0 C: 0
SWC-136	Pass	Unencrypted Private Data On-Chain.	SHILLD.sol	L: 0 C: 0

We scan the contract for additional security issues using MYTHX and industry-standard security scanning tools.

Smart Contract Vulnerability Details

SWC-103 - Floating Pragma.

CWE-664: Improper Control of a Resource Through its Lifetime.

References:

Description:

Contracts should be deployed with the same compiler version and flags that they have been tested with thoroughly. Locking the pragma helps to ensure that contracts do not accidentally get deployed using, for example, an outdated compiler version that might introduce bugs that affect the contract system negatively.

Remediation:

Lock the pragma version and also consider known bugs (<https://github.com/ethereum/solidity/releases>) for the compiler version that is chosen.

Pragma statements can be allowed to float when a contract is intended for consumption by other developers, as in the case with contracts in a library or EthPM package. Otherwise, the developer would need to manually update the pragma in order to compile locally.

References:

Ethereum Smart Contract Best Practices - Lock pragmas to specific compiler version.

Smart Contract Vulnerability Details

SWC-108 - State Variable Default Visibility

CWE-710: Improper Adherence to Coding Standards

Description:

Labeling the visibility explicitly makes it easier to catch incorrect assumptions about who can access the variable.

Remediation:

Variables can be specified as being public, internal or private. Explicitly define visibility for all state variables.

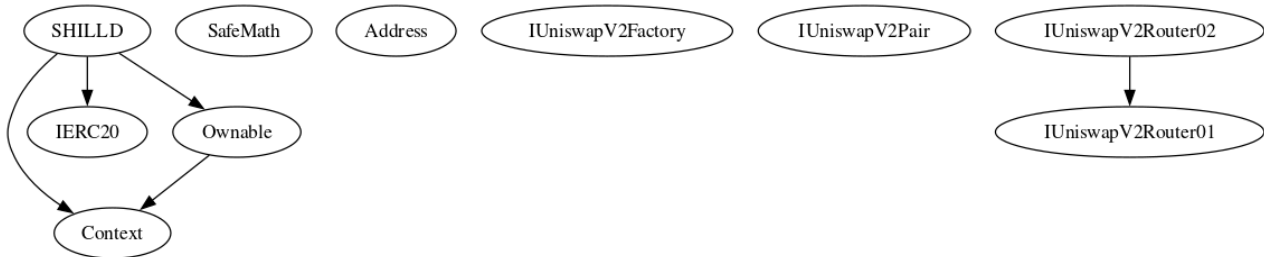
References:

Ethereum Smart Contract Best Practices - Explicitly mark visibility in functions and state variables

Inheritance

The contract for **SHILLD** has the following inheritance structure.

The Project has a Total Supply of 10,000,000



Privileged Functions (onlyOwner)

Please Note if the contract is Renounced none of this functions can be executed.

Function Name	Parameters	Visibility
waiveOwnership		Public
transferOwnership	address newOwner	Public
lock	uint256 time	Public
setMarketPairStatus	address account, bool newValue	Public
setIsTxLLimitExempt	address holder, bool exempt	External
setIsExcludedFromFee	address account, bool newValue	Public
setBuyTaxes	uint256 newMarketingTax	Public
setSellTaxes	uint256 newMarketingTax	External
setMaxTxAmount	uint256 maxTxAmount	External
enableDisableWalletLimit	bool newValue	External
setIsWalletLimitExempt	address holder, bool exempt	External
setWalletLimit	uint256 newLimit	External

Function Name	Parameters	Visibility
setNumTokensBeforeSwap	uint256 newLimit	External
setMarketingWalletAddress	address newAddress	External
setSwapAndLiquifyEnabled	bool _enabled	Public
setSwapAndLiquifyByLimitOnly	bool newValue	Public
changeRouterVersion	address newRouterAddress	Public

Smart Contract Advance Checks

ID	Severity	Name	Result	Status
SHILLD-01	Low	Potential Sandwich Attacks.	Pass	Not-Found
SHILLD-02	Low	Function Visibility Optimization	Pass	Pending
SHILLD-03	High	Lack of Input Validation.	Fail	Pending
SHILLD-04	High	Centralized Risk In addLiquidity.	Pass	Not-Found
SHILLD-05	Low	Missing Event Emission.	Fail	Pending
SHILLD-06	Low	Conformance with Solidity Naming Conventions.	Pass	Not-Found
SHILLD-07	Low	State Variables could be Declared Constant.	Pass	Not-Found
SHILLD-08	Low	Dead Code Elimination.	Pass	Not-Found
SHILLD-09	High	Third Party Dependencies.	Pass	Detected
SHILLD-10	High	Initial Token Distribution.	Pass	Detected
SHILLD-11	High		Pass	Pending
SHILLD-12	High	Centralization Risks In The X Role	Pass	Pending
SHILLD-13	Informational	Extra Gas Cost For User..	Pass	Pending
SHILLD-14	Medium	Unnecessary Use Of SafeMath	Fail	Pending
SHILLD-15	Medium	Symbol Length Limitation due to Solidity Naming Standards.	Pass	Not-Found
SHILLD-16	Medium	Taxes can be up to 100%	Pass	Pending
SHILLD-17	Informational	Conformance to numeric notation best practice.	Pass	Not-Found
SHILLD-18	Informational	Stop Transactions by using Enable Trade.	Pass	Pending

SHILLD-03 | Lack of Input Validation.

Category	Severity	Location	Status
Volatile Code	 High	SHILLD.sol: L: 161 C: 14, L: 180 C: 14, L: 541 C: 14, L: 545 C: 14, L: 549 C: 14, L: 570 C: 14, L: 574 C: 14, L: 578 C: 14, L: 582 C: 14, L: 591 C: 14, L: 596 C: 14, L: 604 C: 14	 Pending

Description

The given input is missing the check for the non-zero address.

The given input is missing the check for the .

Remediation

We advise the client to add the check for the passed-in values to prevent unexpected errors as below:

```
...
    require(receiver != address(0), "Receiver is the zero address");
...
...
    require(value X limitation, "Your not able to do this function");
...
```

We also recommend customer to review the following function that is missing a required validation. .

SHILLD-05 | Missing Event Emission.

Category	Severity	Location	Status
Volatile Code	 Low	SHILLD.sol: L: 541 C: 14, L: 545 C: 14, L: 549 C: 14, L: 553 C: 14, L: 559 C: 14, L: 565 C: 14, L: 570 C: 14, L: 574 C: 14, L: 578 C: 14, L: 582 C: 14, L: 586 C: 14, L: 596 C: 14, L: 608 C: 14	 Pending

Description

Detected missing events for critical arithmetic parameters. There are functions that have no event emitted, so it is difficult to track off-chain changes. The linked code does not create an event for the transfer.

Remediation

Emit an event for critical parameter changes. It is recommended emitting events for the sensitive functions that are controlled by centralization roles.

SHILLD-14 | Unnecessary Use Of SafeMath

Category	Severity	Location	Status
Logical Issue	 Medium	SHILLD.sol: L: 28 C: 9	 Pending

Description

The SafeMath library is used unnecessarily. With Solidity compiler versions 0.8.0 or newer, arithmetic operations

will automatically revert in case of integer overflow or underflow.

library SafeMath {

An implementation of SafeMath library is found.

using SafeMath for uint256;

SafeMath library is used for uint256 type in contract.

Remediation

We advise removing the usage of SafeMath library and using the built-in arithmetic operations provided by the

Solidity programming language

Project Action

Technical Findings Summary

Classification of Risk

Severity	Description
 Critical	Risks are those that impact the safe functioning of a platform and must be addressed before launch. Users should not invest in any project with outstanding critical risks.
 High	Risks can include centralization issues and logical errors. Under specific circumstances, these major risks can lead to loss of funds and/or control of the project.
 Medium	Risks may not pose a direct risk to users' funds, but they can affect the overall functioning of a platform
 Low	Risks can be any of the above but on a smaller scale. They generally do not compromise the overall integrity of the Project, but they may be less efficient than other solutions.
 Informational	Errors are often recommended to improve the code's style or certain operations to fall within industry best practices. They usually do not affect the overall functioning of the code.

Findings

Severity	Found	Pending	Resolved
 Critical	0	0	0
 High	0	0	0
 Medium	1	0	0
 Low	2	0	0
 Informational	0	0	0
Total	3	0	0

Social Media Checks

Social Media	URL	Result
Twitter	https://twitter.com/get_shild	Pass
Other		Fail
Website	https://shild.io/	Pass
Telegram	https://t.me/shild_official	Pass

We recommend to have 3 or more social media sources including a completed working websites.

Social Media Information Notes:

Auditor Notes: undefined

Project Owner Notes:



Assessment Results

Score Results

Review	Score
Overall Score	91/100
Auditor Score	85/100
Review by Section	Score
Manual Scan Score	28/53
SWC Scan Score	33 /37
Advance Check Score	30 /19

The Following Score System Has been Added to this page to help understand the value of the audit, the maximum score is 100, however to attain that value the project must pass and provide all the data needed for the assessment. Our Passing Score has been changed to 80 Points, if a project does not attain 80% is an automatic failure. Read our notes and final assessment below.

Audit Passed



Assessment Results

Important Notes:

- No issues or vulnerabilities were found.

Auditor Score =85
Audit Passed



Appendix

Finding Categories

Centralization / Privilege

Centralization / Privilege findings refer to either feature logic or implementation of components that act against the nature of decentralization, such as explicit ownership or specialized access roles in combination with a mechanism to relocate funds.

Gas Optimization

Gas Optimization findings do not affect the functionality of the code but generate different, more optimal EVM opcodes resulting in a reduction on the total gas cost of a transaction.

Logical Issue

Logical Issue findings detail a fault in the logic of the linked code, such as an incorrect notion on how block.timestamp works.

Control Flow

Control Flow findings concern the access control imposed on functions, such as owner-only functions being invoke-able by anyone under certain circumstances.

Volatile Code

Volatile Code findings refer to segments of code that behave unexpectedly on certain edge cases that may result in a vulnerability.

Coding Style

Coding Style findings usually do not affect the generated byte-code but rather comment on how to make the codebase more legible and, as a result, easily maintainable.

Inconsistency

Inconsistency findings refer to functions that should seemingly behave similarly yet contain different code, such as a constructor assignment imposing different requirements on the input variables than a setter function.

Coding Best Practices

ERC 20 Coding Standards are a set of rules that each developer should follow to ensure the code meets a set of criteria and is readable by all the developers.

Disclaimer

Assure Defi has conducted an independent security assessment to verify the integrity of and highlight any vulnerabilities or errors, intentional or unintentional, that may be present in the reviewed code for the scope of this assessment. This report does not constitute agreement, acceptance, or advocacy for the Project, and users relying on this report should not consider this as having any merit for financial advice in any shape, form, or nature. The contracts audited do not account for any economic developments that the Project in question may pursue, and the veracity of the findings thus presented in this report relate solely to the proficiency, competence, aptitude, and discretion of our independent auditors, who make no guarantees nor assurance that the contracts are entirely free of exploits, bugs, vulnerabilities or deprecation of technologies.

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